

[Title: What does topological field theory require of probabilistic theories?]

Pedro Lauand¹

¹*Perimeter Institute for Theoretical Physics, Waterloo, Ontario, Canada*

[Abstract.]

INTRODUCTION

Quantum theory is one of the most empirically successful theories in science. It describes its experiments probabilistically: the theory assigns probabilities to the possible outcomes of the measurements one can perform upon physical systems. Every empirical claim of the theory therefore is described by collecting measurement statistics of repeated experimental runs. Despite its success, the foundational aspects of its mathematical formulation have remained elusive [cite]. In its textbook presentation, quantum theory (QT) is expressed through complex vectors in a Hilbert space and self-adjoint operators—an abstract structure whose postulates arguably carry no direct physical motivation. Over the last decades, however, this description has been reformulated into an *operational view*, i.e. to consider as its primitive notions “self-evident” features of general laboratory situations and derive the abstract formalism from simple premises over these primitive notions. This shift to the operational perspective allows one not to refer to the specific mathematical framework of QT, but rather understand QT as a consequence of natural information-theoretic constraints over the mathematical framework of Generalized Probabilistic Theories (GPTs) [1–3] in which QT can be situated as one theory among many others [4–6].

Based on these ideas, the next wave of axiomatizations of QT, also known as *reconstructions* of QT, started. The most famous historical example that is often cited as a role model for reconstructions of QT is special relativity. In this example, Einstein showed that the Lorentz transformations can be derived from essentially two simple physical principles: the constancy of the speed of light and the relativity principle. This was an important step towards improving our conceptual understanding of the Lorentz transformations. Reconstructions of QT achieve a similar goal: by considering information-processing assumptions, such reconstructions try to make the abstract formalism of QT more physically motivated. The first reconstruction of this flavor was the seminal work of Hardy [4], which later inspired new solutions to the reconstruction problem [5, 6]. Each of these reconstructions selects quantum theory out of the landscape of GPTs as the unique GPT that satisfies a set of simple principles.

The most famous physical principle in GPTs is the *no-signalling* principle [7], which can be understood as the weakest condition any “reasonable” physical theory must satisfy, in the sense of being compatible with special relativity [8].

The no-signalling principle can be formulated as the requirement that operations on separate systems commute [1]. This condition guarantees that when two devices operate at spacelike separated locations, their correlations respect relativistic causality. In this sense, the no-signalling principle can be understood as the compatibility of relativistic causality with the operational theory.

The no-signalling principle considers the kinematical picture of relativity: systems are embedded as points of spacetime, and points held at spacelike separation cannot instantaneously influence each other’s measurement statistics. The only feature of spacetime this invokes is the causal relation between a pair of points. This suggests a question more precise than the compatibility of a theory with relativistic causality: what does a probabilistic theory require in order to be distributed over space?

[B6]

[Warning]

[B7]

[B8]

[Declare results.]

-
- [1] J. Barrett, Information processing in generalized probabilistic theories, *Phys. Rev. A* **75**, 032304 (2007), [arXiv:quant-ph/0508211](#).
 - [2] M. P. Müller, Probabilistic theories and reconstructions of quantum theory, *SciPost Phys. Lect. Notes* **28** (2021), [arXiv:2011.01286](#).
 - [3] M. Plávala, General probabilistic theories: An introduction, *Phys. Rep.* **1033**, 1 (2023), [arXiv:2103.07469](#).
 - [4] L. Hardy, Quantum theory from five reasonable axioms (2001), [arXiv:quant-ph/0101012](#).
 - [5] G. Chiribella, G. M. D’Ariano, and P. Perinotti, Informational derivation of quantum theory, *Phys. Rev. A* **84**, 012311 (2011), [arXiv:1011.6451](#).
 - [6] L. Masanes and M. P. Müller, A derivation of quantum theory from physical requirements, *New J. Phys.* **13**, 063001 (2011), [arXiv:1004.1483](#).
 - [7] S. Popescu and D. Rohrlich, Quantum nonlocality as an axiom, *Found. Phys.* **24**, 379 (1994).
 - [8] N. Gisin, J.-D. Bancal, Y. Cai, P. Remy, A. Tavakoli, E. Zambini Cruzeiro, S. Popescu, and N. Brunner, Constraints on nonlocality in networks from no-signaling and independence, *Nat. Commun.* **11**, 2378 (2020), [arXiv:1906.06495](#).